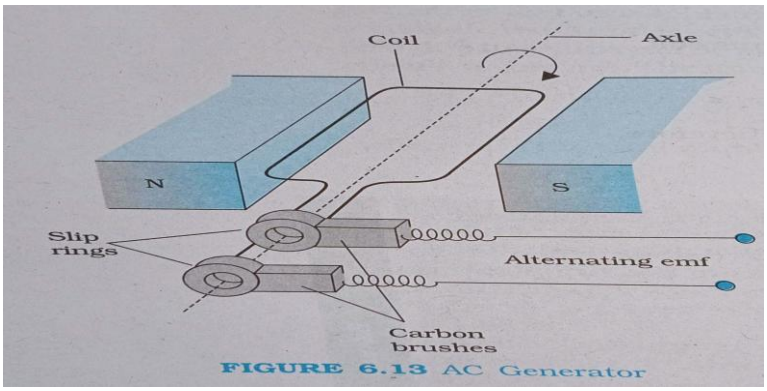
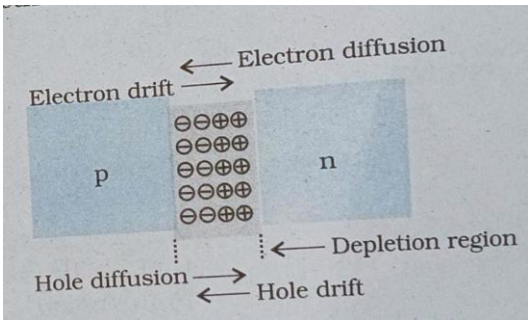
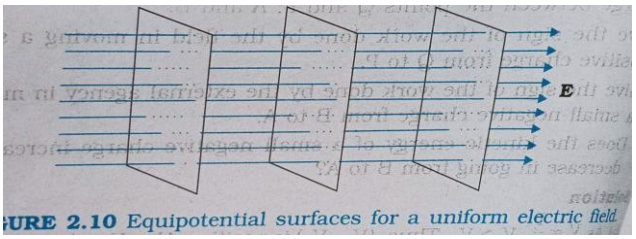
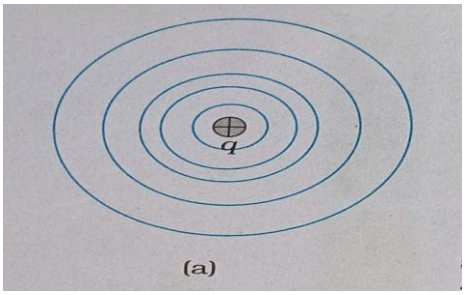
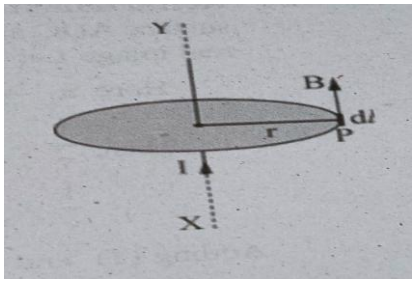
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD

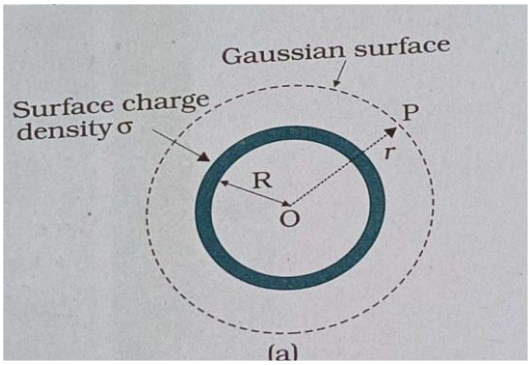
II PUC EXAM- 2 APRIL/MAY- 2025

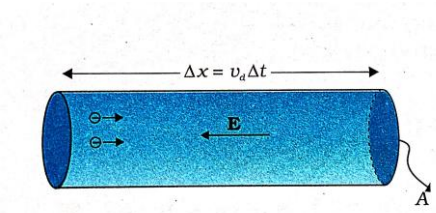
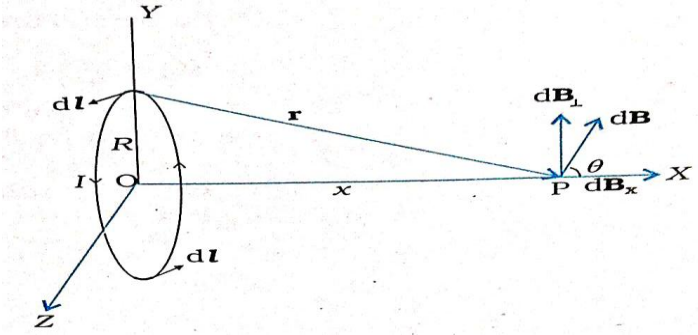
SUBJECT: 33- PHYSICS SCHEME OF EVALUATION MAX MARKS: 70

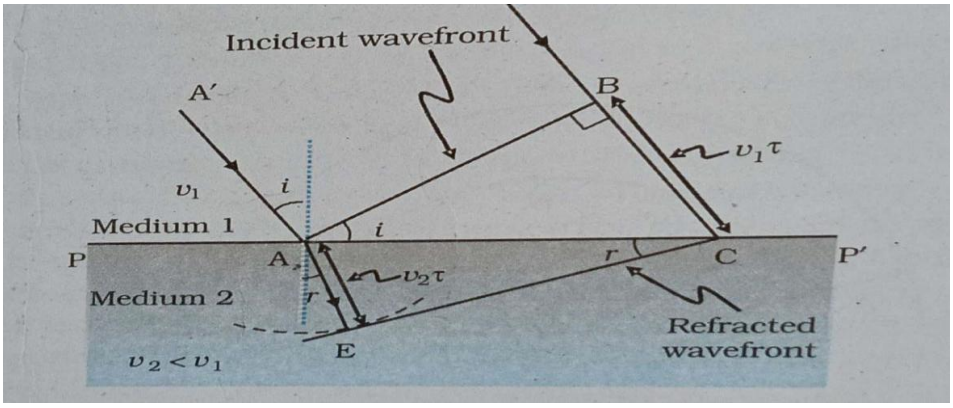
Q No	KEY ANSWERS	MARKS
I	**Pick the correct option among the four given options for ALL the following**	**15 x1=15**
1	c) F α $\frac{1}{r^2}$	**1**
2	b) dielectric strength	**1**
3	c)current density	**1**
4	b) low resistance in parallel	**1**
5	a) diamagnetic	**1**
6	c)does not induces current in the ring	**1**
7	a) mutual induction	**1**
8	d) J C Maxwell	**1**
9	c) f = $\frac{R}{2}$	**1**
10	d)both statement I and statement II are correct	**1**
11	b) polarization	**1**
12	d) thermionic emission	**1**
13	a) Hydrogen atom	**1**
14	a) a-ii, b- iii, c-i	**1**
15	b) trivalent	
II	Fill in the blanks by choosing appropriate answer given in the bracket for ALL of the following	**5 x 1 =5**
16	linear	**1**
17	Electrostatic potential	**1**
18	iron	**1**
19	zero	**1**
20	refraction	**1**
III	**PART - B** **Answer any FIVE of the following questions**	**5 x 2 = 10**
21	* Electric field lines start from positive charges and ends at negative charges. * Two field lines can never cross each other. * Electrostatic field lines do not from any closed loop * In a charge- free region, electric field lines can be taken to be continuous curves without any breaks. (any two)	**1+1**

22	F= E q F = 0.8 x2 x10^{-6} $= 1.6x10^{-6}N$	**1** **1**
23	Junction rule: In an electrical network, at any junction, the sum of the currents entering the junction is equal to the sum of currents leaving the junction. Loop rule: in an electrical network, the algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.	**1** **1**
24	Current sensitivity of galvanometer depends on * Number of turns in the coil (N) * Area of the coil (A) * Magnetic field used (B) * torsional constant of the spring (k) (any two)	**1+1**
25	 FIGURE 6.13 AC Generator Diagram (1 mark) label any two parts (1 mark)	**1+1**
26	$P_{av} = V_{rms}I_{rms}cos\varphi$ V_{rms} rms value of voltage I_{rms} rms value of current φ phase difference between voltage and current	**1** **1**
27	*Infrared detectors are used in Earth satellites. * Electric devices emit infrared rays and widely used in remote switches of TV sets, video recorders and hi-fi systems. * They are used to trap by greenhouse gases such as carbon dioxide and water vapour.	**1+1**
28 a)	 c) The processes of conversion of A C in to D C is called rectification	**1** **1**

IV	**PART - C** Answer any FIVE of the following questions	**5 x 3 = 15**
29	a) The surface having same potential at all points is called equi-potential surface  URE 2.10 Equipotential surfaces for a uniform electric field For a point charge  	**1** **1** **1**
30	Consider a straight infinite current carrying conductor, having I. Let p be a point at a distance r as shown in figure  According to Ampere`s circuital law $\oint \overrightarrow{B} \cdot \overrightarrow{dl} = \mu_0 I$ B $(2\pi r) = \mu_0 I$ $B = \frac{\mu_0 I}{2\pi r}$	**1** **1** **1**
31	Magnetic susceptibility is the ratio of magnetization to magnetic intensity or $\chi = \frac{M}{H}$ Magnetic relative permeability is the ratio of permeability of medium to the permeability of free space. Or $\mu_r = \frac{\mu}{\mu_0}$ $\mu_r = 1 + \chi$	**1** **1** **1**
32	a) The phenomenon in which emf induces in a coil due to change in magnetic flux linked with the coil is called electromagnetic induction. b) The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit. $e = -\frac{d\varphi}{dt}$ $\frac{d\varphi}{dt}$ → rate of change of magnetic flux	**1** **1** **1**

33	Given L =1.0 x 10^{-3} H, C = 1.0 x10^{-9}F $\omega = \frac{1}{\sqrt{LC}}$ $\omega = \frac{1}{\sqrt{1.0x10^{-3}x1.0x10^{-9}}}$ $\omega = 10^6$ rad/s	**1** **1** **1**
34	* Light must passes from denser to rarer medium. * Angle of incidence must be greater than critical angle. Technological applications, TIR is used * In total reflecting prisms * In optical fibres. (Any one)	**1+1** **1**
35	Consider a hydrogen atom, in which an electron of mass m and charge e is revolving around the nucleus in a circular stationary orbit of radius r. The electrostatic potential energy of the electron, $U = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$ Kinetic energy of the electron $K = \frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$ Total energy of electron is E = U + K $E = -\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$	**1** **1** **1**
36	The process of combination of two or more lighter nuclei in to a heavy nucleus is called nuclear fusion. $4\ {}_1^1H + 2e^- \rightarrow {}_2^4He + 2v + 6\gamma + 26.7\ MeV$ Or explanation in words. This nuclear reaction takes place at very high temperature.	**1** **1** **1**
V	**PART - D** Answer any THREE of the following questions	**3x 5=15**
37	Deduce an expression for electric field at a point outside a thin charged spherical shell using Gauss`s law. Consider a thin charged spherical shell of radius R. Let q be the charge on the shell. Let p be a point outside the shell at a distance r from its centre. To find electric flux consider a Gaussian spherical surface of radius r round the shell as shown in figure.  (a)	**1** **1**

	Electric flux due to a small area ds is dφ = E cosθ ds But θ = 0 dφ = E ds Now Electric flux due to Gaussian surface is $\Phi = E \ 4\pi r^2$ ----------(1) According to Gauss`s law of electrostatics $\Phi = \frac{1}{\varepsilon_0} q$ ----------(2) From equations (1) and (2) $E \ 4\pi r^2 = \frac{1}{\varepsilon_0} q$ $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$	**1** **1** **1**
38	Obtain an expression for conductivity of conducting electrons in a metallic conductor in terms of charge (e) and mass (m) of the electrons. Consider a conductor of length L and area of cross section 'A'. let 'E' be the electric field over the conductor. The free electrons constitute electric current I through the conductor.  The expression for drift velocity in terms of current is given by, $v_d = \frac{I}{neA}$ ---------- (1) Where 'n' is electron density, 'e' charge of electron Also $v_d = \frac{eE\tau}{m}$ ---------- (2) Where τ relaxation time of electrons, m is mass of electron. From equations (1) and (2) $\frac{I}{neA} = \frac{eE\tau}{m}$ $\frac{I}{A} = \frac{e^2 n\tau \ E}{m}$ J = σ E Where σ is conductivity and $\sigma = \frac{e^2 n\tau}{m}$ J is current density	**1** **1** **1** **1** **1**
39	Derive expression for magnetic field at a point on axis of a circular current loop.  	**1**

	Consider a circular current loop of radius R having current I in anticlockwise direction as shown in figure. Let p be a point at a distance x from the centre of loop. The magnetic field due to current element at point p is given by according to Biot - Savart`s law $$dB = \frac{\mu_0}{4\pi}\frac{I\,dl\,sin\theta}{r^2}$$ But $\theta = 90^0$ And $dB = \frac{\mu_0}{4\pi}\frac{I\,dl}{r^2}$----------(1) The direction of dB is as shown in figure The net magnetic field due to the loop at point p is obtain by resolving and addition of components, $B = \sum dB\ cos\alpha$ $B = \frac{\mu_0}{4\pi}\frac{I\sum dl\ cos\alpha}{r^2}$ ----------(2) But $\sum dl = 2\pi R$ and $cos\alpha = \frac{R}{r}$ Now $B = \frac{\mu_0}{4\pi}\frac{I2\pi R}{r^2}\ (\frac{R}{r})$ $B = \frac{\mu_0}{4\pi}\frac{I2\pi R^2}{r^3}$ $B = \frac{\mu_0}{4\pi}\frac{2\pi IR^2}{\sqrt[3]{x^2+R^2}}$	**1** **1** **1** **1**
40	a) Prove Snell`s law of refraction of light using plane wave front passes from rarer to denser medium. Consider a plane wave front passing from denser to rarer medium as shown in figure. Let AB be the incident wave front, and EC be refracted wave front  In triangle ABC sin i $= \frac{BC}{AC} = \frac{v_1\tau}{AC}$ ---------- (1) and In triangle AEC sin r $= \frac{AE}{AC} = \frac{v_2\tau}{AC}$----------(2) From (1) and (2) $\frac{sin\,i}{sin\,r} = \frac{v_1\tau}{v_2\tau} = \frac{v_1}{v_2}$ = constant b) uses of polariods * They are used to control the intensity of light, in sun glasses and windowpanes. etc. * They are used in photographic cameras and 3D movie cameras.	**1** **1** **1** **1** **1+1**

41 a) Explain classification of solids using band theory

Solids are classified as conductors, semiconductors and insulators using energy band theory

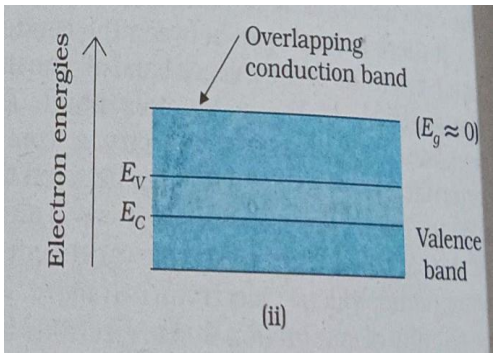
In conductors valance band and conduction bands overlapped. **1**

In semiconductors there is small energy gap (< 3 eV) between valance band and conduction band. **1**

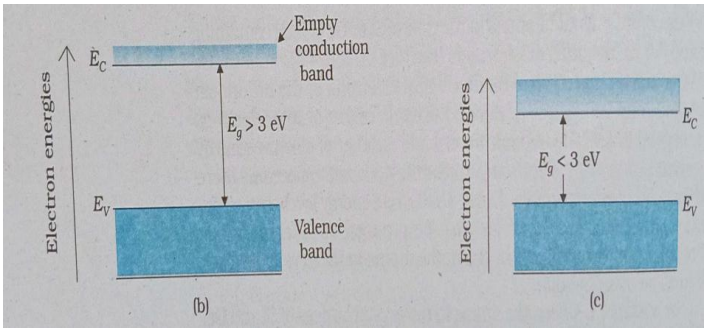
In insulators there is a large energy gap (> 3 eV) between valance band and conduction band **1**

OR

With labeled diagram as follows



i) Conductors



1+1

ii)insulators iii) semiconductors

Intrinsic semiconductors:

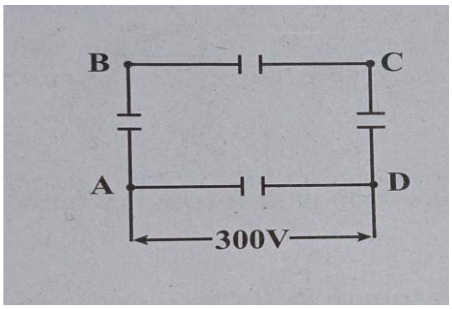
* These are pure semiconductors.

* They have equal number of free electrons and holes.

Extrinsic semiconductors:

* These are impure semiconductors

* They have unequal number of free electrons and holes.

VI	Answer any TWO of the following	2 x 5 = 10
42		
	Taking Three capacitors in series across adjacent corners,	
	$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$	**1**
	$\frac{1}{C_S} = \frac{1}{9} + \frac{1}{9} + \frac{1}{9}$	**1**
	$C_S = 3\mu F$	
	Effective capacitance (network) across adjacent corners given by	
	$C_{eff} = C_S + C_4$	**1**
	C_{eff}= 3 + 9 =12μF	**1**
	Charge on each in series combination is	
	Q =$C_S V$ = 3 $x10^{-6} x300$ = 900μC	**1**
	Charge on C_4 capacitor is	
	Q = C_4 V = $9x10^{-6}$ $x300$ = 2700μC	
43	$\varepsilon_{\text{eff}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$	**1**
	$\varepsilon_{eff} = \frac{8x2+10x1}{1+2} = \frac{26}{3} V$	**1**
	$r_{eff} = \frac{r_1 r_2}{r_1 + r_2} = \frac{1x2}{1+2} = \frac{2}{3}\Omega$	**1**
	$V = \varepsilon_{eff} - I r_{eff}$	
	$IxR = Ix20 = \frac{26}{3} - \frac{I\,x2}{3}$	
	$I = \frac{26}{62} = 0.42A$	**1**
	$V = I\,R = 0.42X20 = 8.4volt$	**1**
44	$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	**1**
	$\frac{1}{v} = \frac{1}{14} - \frac{1}{21} = \frac{1}{42}$	**1**
	$v = 42\ cm$	**1**
	$m = \frac{42}{-21} = -2$	**1**
	When the object is moved away from the lens the height of the image decreases.	**1**

45	Incident energy of photon, $E_I = h\,\gamma$	**1**
	$= \frac{6.63x10^{-34}x6.50x10^{14}}{1.6x10^{-19}}$	**1**
	$E_I = 2.69\ eV$	
	Kinetic energy of photo electrons $k_{max} = E_I - \varphi$	**1**
	$k_{max} = 2.69 - 2.20 = 0.49eV$	**1**
	Stopping potential of photo electrons, $V_0 = 0.49\ volt$	**1**

(For numerical problems any alternative method with formula and detailed correct answers, marks can be awarded)